

# Sravanthi Chede



## Contact & Info

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📍 Hyderabad, India    🌐 <https://sravanthi-ch.github.io/webpage/>

**Languages:** English, Hindi, Telugu      **DOB:** Nov, 1998

## EDUCATION

|                        |                                                                                                                                                                                                                                                                                                    |
|------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Sept, 2020 - Dec, 2025 | <b>PhD Degree (8.55 / 10)</b> 📍 Punjab<br><i>in Theoretical Computer Science</i><br><i>Indian Institute of Technology (IIT) Ropar</i><br>[Supervisor: Dr. Anil Shukla]<br>Completed the Thesis Defense Seminar, and published two conference papers, one journal paper and a dataset in the field. |
| 2016-2020              | <b>Bachelor Degree (8.24 / 10)</b> 📍 Nagpur<br><i>in Computer Science &amp; Engineering</i><br><i>Indian Institute of Information Technology (IIIT) Nagpur</i><br>Completed 40+ courses, took-up a semester long paid internship on ML and created a bug tracking system for the BTech project.    |
| 2014-2016              | <b>Higher secondary education (96%)</b> 📍 Hyderabad<br><i>in TSBIE board, Excel Junior College</i>                                                                                                                                                                                                 |
| 2014                   | <b>Matriculation (91%)</b> 📍 Hyderabad<br><i>in ICSE board, St. Joseph's Public School</i>                                                                                                                                                                                                         |

## WORK EXPERIENCE

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|-----------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Jan 2026 - Today      | <b>Research Associate [\$]</b> 📍 Chennai<br><i>The Institute of Mathematical Sciences (IMSc)</i><br>Under Prof. Meena Mahajan [ANRF J.C. Bose Fellowship grant]                                                                                                                                                                                                                                                  |
| Sept 2021 - Sept 2025 | <b>Teaching Assistant for BTech students [\$]</b> 📍 Punjab<br><i>Indian Institute of Technology (IIT) Ropar</i><br>Feb-May'21: Introduction to computing and Data Structure<br>July-Dec of 2021, 2022, 2023, 2024: Data Structures<br>Jan-May of 2022, 2023, 2024, 2025: Theory of Computation<br>July-Nov of 2025: Design and Analysis of Algorithms<br>Also guided 40+ BTech student-teams with their projects |
| July 2019 - Dec 2019  | <b>Machine Learning Intern [\$]</b> 📍 Pune<br><i>Tata Research Development and Design Centre (TRDDC)</i><br>The goal of this intern was to assist in Video & Audio analysis for Emotion-Detection using ML & DeepLearning.                                                                                                                                                                                       |
| May 2018 - July 2018  | <b>Intern Software Developer</b> 📍 Hyderabad<br><i>Defence Research Development Organization (DRDO)</i><br>The goal of the project was to develop a website for internal asset management for DOI and a mini-MySQL implementation.                                                                                                                                                                               |

## PUBLICATIONS

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|---------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Journal Article</b><br>2026        | <b>Extending Merge Resolution to a Family of QBF-Proof Systems</b> , Sravanthi Chede & Anil Shukla, <i>ACM Transactions on Computation Theory (ACM TOCT)</i> [Invited Paper], <a href="#">doi</a>                                                     |
| <b>Dataset</b><br>2025                | <b>Hamming weight based model counting benchmarks</b> , Anil Shukla & Sravanthi Chede, <i>Zenodo</i> , <a href="#">doi</a>                                                                                                                            |
| <b>Conference Proceedings</b><br>2024 | <b>Circuits, Proofs and Propositional Model Counting</b> , Sravanthi Chede, Leroy Chew, Anil Shukla, <i>44th IARCS Annual Conference on Foundations of Software Technology and Theoretical Computer Science (FSTTCS), India</i> , <a href="#">doi</a> |
| <b>Conference Proceedings</b><br>2023 | <b>Extending Merge Resolution to a Family of QBF-Proof Systems</b> , Sravanthi Chede & Anil Shukla, <i>40th International Symposium on Theoretical Aspects of Computer Science (STACS), Germany</i> , <a href="#">doi</a>                             |
| <b>Journal Article</b><br>2021        | <b>World War III analysis using signed social networks</b> , Ranjana Roy Chowdhury, Shivam Gupta & Sravanthi Chede, <i>Social Network Analysis and Mining</i> , <a href="#">doi</a>                                                                   |

## Research Accomplishments

- Presented the FSTTCS-24 result as a **poster** at the SYNTACS conference of IIT Ropar in March 2025.
- Delivered an **invited talk** on the FSTTCS-24 result at the Advances in Modern Complexity Theory Workshop 2025 held at IMSc, Chennai.
- An earlier version of the FSTTCS-24 result was accepted as a **workshop paper** in MCW 2024 held at TRDDC, Pune.
- Following the STACS-23 result, an **extended version was invited** to be submitted to the reputed ACM Transactions of Computation Theory Journal.
- Won the '**Best Presenter Award**' at the Research Scholar's day of IIT Ropar in April 2023.
- Gave a **talk** on "Motivation to Proof Complexity" at IIT Ropar as a part of the Departmental Lunch Presentations in March 2023.
- Presented the STACS-23 result as a **poster** at IIT Ropar as a part of the CSE Sankalanam programme in Feb 2023.

## Preprint manuscripts

- **On Proof Systems for #QBF**  
Sravanthi Chede, Leroy Chew, Vaibhav Krishan and Anil Shukla.
- **Understanding Nullstellensatz for QBFs**, [doi](#)  
Sravanthi Chede, Leroy Chew, Balesh Kumar and Anil Shukla.
- **Does QRAT simulate IR-calc? QRAT simulation algorithm for  $\forall\text{Exp}+\text{Res}$  cannot be lifted to IR-calc**, [doi](#)  
Sravanthi Chede and Anil Shukla.

## Research service

- ★ Acted as a sub-reviewer for the SAT-2026 conference.
- ★ Acted as a sub-reviewer for the SAT-2025 conference.
- ★ Active reviewer for the Logical Methods in Computer Science Journal.
- ★ Provided benchmarks for the Model Counting Competition of 2025.

## CAMPUS ACTIVITIES

- PhD Representative for the CSE Dept of IIT Ropar (2022-2024).
- Headed the Treasury department for Fresher's day in 2023 and again in 2024 at CSE, IIT Ropar.
- Was the Technical General Secretary of IIIT Nagpur and organized a national level technical fest 'TANTRAFIESTA' in 2018.
- Head of Research Club, 'REAP' at IIIT Nagpur (2018-2020).
- Headed Institute level 'Anti-Ragging Committee' (2018-2020).
- Headed the Anchoring department in the cultural fest 'ABHIVYAKTI' 2018 at IIIT Nagpur.
- Bagged the First Prize in the Fashion show 'INCREDIBLE INDIA' of ABHIVYAKTI 2017 at IIIT Nagpur.

## Other Interests

Books 📖

Movies 🎬

Driving 🚗

Classical dance 🎭

Travels 🌍

Singing 🎤